

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 37)

Suggested Solution

Paper 1

1. B	11. A	21. B	31. D	41. D
2. B	12. D	22. C	32. B	42. C
3. B	13. D	23. D	33. B	43. C
4. D	14. B	24. C	34. D	44. B
5. A	15. A	25. A	35. B	45. D
6. A	16. C	26. A	36. C	
7. C	17. B	27. B	37. D	
8. A	18. C	28. D	38. A	
9. B	19. C	29. A	39. A	
10. C	20. A	30. C	40. C	

Explanation

1. Answer: B

(1) is correct. The monetary cost of renting an industrial building unit to make short films will decrease.

(3) is correct. This will reduce the value of the highest-valued option forgone (i.e. making short films at home) and the cost of renting an industrial building unit to make short films will decrease.

(2) is incorrect. This will reduce the value of the chosen option, without affecting the highest-valued option forgone.

2. Answer: B

(1) and (3) are correct. They do not involve a value judgement and are refutable by facts, so they are positive statements.

(2) is incorrect. Whether Hong Kong International Airport should be further expanded involves a value judgement.

(4) is incorrect. Which airport is more beautiful involves a value judgement.

3. Answer: B

Opening most leisure facilities under the LCSD for free use on 'Sport For All Day 2024' is about who can use the facilities while allocating the sessions for free use of facilities by drawing lots is about how to distribute the facilities to people in society. Therefore, the case is related to the 'for whom to produce' problem.

Option D is incorrect. The types and quantities of facilities provided by the LCSD on the 'Sport For All Day 2024' did not change, so it is not related to the 'what to produce' question.

4. Answer: D

5. Answer: A

Option B is incorrect. These free soft drinks are economic goods. However, a good is considered an economic good if its quantity available is not sufficient to satisfy all human wants.

Option D is incorrect. These free soft drinks are private goods which are excludable and rival in consumption.

6. Answer: A

7. Answer: C

	Firm A	Firm B
Average labour productivity	4,000 units / (10 × 50) man-hours = 8 units/man-hour	4,800 units / (8 × 60) man-hours = 10 units/man-hour

According to the above table, firm A's average labour productivity is lower than that of firm B.

(3) is correct. Firm A's average labour productivity is lower and its percentage of output with quality up to standard is higher, implying that firm A's workers may not have to rush for more output to earn more income, so firm A is more likely to pay its workers by time rate. Firm B's average labour productivity is higher but its percentage of output with quality up to standard is lower, implying that its workers may rush for more output at the expense of quality, so firm B is more likely to pay its workers by piece rate.

8. Answer: A

Labour (units)	Machines (units)	Average product (units)	Total product (units)	Marginal product (units)
5	3	2	6	-
5	4	4	16	10
5	5	7	35	19
5	6	X must be smaller than 9 (= 54 / 6)	must be smaller than 54 (= 35 + 19)	must be smaller than 19

Given the data illustrate the law of diminishing marginal returns, the marginal product of machines begins to diminish when the 6th unit of machines is employed. The marginal product of the 6th unit of machines must be smaller than 19 units. The total product must be smaller than 54 units and the average product (X) must be smaller than 9 units.

9. Answer: B

10. Answer: C

Output (units)	1	2	3	4
Average variable cost (\$)	3	4	5	6
Total variable cost (\$)	3	8	15	24
Total fixed cost (\$)	3	3	3	3
Total cost (\$)	6	11	18	27
Marginal cost (\$)	3	5	7	9

Producer surplus = Total revenue – Total variable cost = $(\$7 \times 3) - \$15 = \$6$

At the profit-maximising output level (3 units), marginal revenue equals marginal cost (\$7).

Profit = Total revenue – Total cost = $(\$7 \times 3) - \$18 = \$3$

11. Answer: A

There is an excess demand for newly constructed HOS flats, so the quantity supplied is smaller than the quantity demanded and the selling price is lower than the equilibrium price.

12. Answer: D

When the government of Country A increases the quota for imported foreign workers in the catering industry, the equilibrium wage rate for workers in the catering industry will decrease. The production cost of the catering industry will decrease and supply will increase. The supply curve will shift to the right from S_0 to S_2 .

More tourists visiting Country A will lead to an increase in demand for catering services. The demand curve will shift to the right from D_0 to D_2 .

Therefore, the new equilibrium point will be at E_4 .

13. Answer: D

Physical therapists are factors of production for the provision of physical therapy services, so the demand for physical therapists can be regarded as a derived demand for physical therapy services.

Option A is incorrect. Doctors are factors of production for the provision of medical services, so the demand for doctors is a derived demand for medical services.

Option B is incorrect. Crocodile meat and crocodile skin are in joint supply as they are both products from crocodiles.

Option C is incorrect. Tablets and network services are usually used together to satisfy the need for using tablets, so they are in joint demand.

14. Answer: B

(1) is correct. Given instant noodles are inferior goods to Country A's citizens, when the income of Country A's citizens in general increases, the demand for instant noodles will decrease. Under the effective price floor, the marginal benefit from consuming the last unit of instant noodles will remain unchanged (at the level of the price floor).

(2) is correct. Under the effective price floor, the quantity supplied is greater than the quantity demanded and there is an excess supply. When the demand for instant noodles decreases, the quantity demanded will decrease, thereby widening the gap between the quantity supplied and the quantity demanded, i.e. there will be an increase in excess supply.

(3) is incorrect. When the demand for instant noodles decreases, the price of instant noodles will remain unchanged at the price floor while the quantity transacted will decrease. Therefore, total expenditure will decrease.

15. Answer: A

Total subsidy = Per-unit subsidy $\times$ Quantity transacted after the subsidy

When the per-unit subsidy on good X is halved and the total subsidy granted by the government is also halved, this implies that the quantity transacted remains unchanged.

Option A is correct. If the elasticity of demand for good X is equal to 0 (i.e. demand for good X is perfectly inelastic), the quantity transacted will remain unchanged when the per-unit subsidy is reduced.

Options B, C and D are incorrect. The quantity transacted will decrease when the per-unit subsidy is halved in all three situations.

16. Answer: C

Q_3 is an ineffective quota, while Q_1 is an effective quota.

When the quota is lowered, the quantity transacted will decrease from Q_2 to Q_1 and the market price will increase from P_2 to P_3 . The new consumers' total expenditure is $(P_3 \times Q_1)$ while the original consumers' total expenditure is $(P_2 \times Q_2)$. Consumers' total expenditure will decrease by $(P_2 \times Q_2) - (P_3 \times Q_1)$.

(1) is incorrect. After the imposition of the effective quota, the supply curve will be kinked and become vertical at Q_1 . At Q_1 , quantity supplied equals quantity demanded and the market is at equilibrium.

17. Answer: B

The imposition of a per-unit sales tax will cause the supply to decrease and the price of good X will increase. The decrease in the price of the substitute for good X will cause the demand for good X to decrease and the price of good X will decrease. The effect of the imposition of a per-unit sales tax on the price is offset by the effect of the decrease in demand on the price.

(2) is correct. As both the supply of and demand for good X decrease, the quantity transacted will decrease.

(1) is incorrect. Since the quantity transacted of good X decreases, i.e. to fewer than 40 units, the tax revenue must be smaller than \$160.

(3) is incorrect. At the new equilibrium, marginal benefit is higher than marginal cost. There is a deadweight loss.

18. Answer: C

19. Answer: C

The price ceiling changes from ineffective (P_1) to effective (P_2). The price will decrease from the equilibrium price to P_2 , and the quantity transacted will decrease. Since the quantity of output under the effective price ceiling is below the efficient output level, there will be a deadweight loss.

(1) is correct. As the quantity transacted and the price decrease, the producer surplus will decrease.

(2) is correct. Under the effective price ceiling, there will be a deadweight loss, so total social surplus will decrease.

(4) is correct. As the quantity transacted and the price decrease, total sales revenue will decrease.

20. Answer: A

(1) is correct. After a per-unit sales tax is imposed, the quantity transacted will decrease and the price received by producers will also decrease. Producer surplus will decrease. As the quantity of output is below the efficient output level, a deadweight loss is created in the market.

(3) is correct. After an effective price ceiling is imposed, both the price and quantity transacted will decrease. Producer surplus will decrease. As the quantity of output is below the efficient output level, a deadweight loss is created in the market.

(2) is incorrect. After an effective quota is imposed, the price will increase and the quantity transacted will decrease. Producer surplus may increase, decrease or remain unchanged, depending on the price elasticity of demand.

(4) is incorrect. A decrease in demand will cause the price and quantity transacted to decrease, and producer surplus will decrease. However, as marginal benefit equals marginal cost at the quantity transacted, there will be no deadweight loss.

21. Answer: B

If the shopping mall gives compensation to the affected residents, the external cost can be internalised, reducing the divergence between private and social costs.

Option A is incorrect. If the shopping mall gives compensation to the affected residents, this money will increase the marginal private cost of holding outdoor concerts. The quantity of noise created by the shopping mall will be reduced to a level closer to the socially optimal quantity. However, noise pollution may still exist.

Option C is incorrect. Even if the shopping mall has the right to hold outdoor concerts on its rooftop, the marginal social cost is still higher than the marginal private cost, so the divergence between private and social costs still exists.

Option D is incorrect. As marginal social cost is higher than marginal social benefit at the equilibrium quantity, from the viewpoint of society, there is over-production of music, instead of under-production.

22. Answer: C

(2) is correct. People living in public rental housing units generally have low income. If the government increases the rent of public rental housing units, their income after taxes and social transfers will decrease while people with high income will not be affected. The income gap between the rich and the poor will widen. The Gini coefficient based on post-tax post-social transfer monthly household income will increase.

(3) is correct. Decreasing the marginal tax rates for the salaries tax will allow high-income earners to pay less tax while low-income earners will not be affected. The income gap between the rich and the poor will widen and the Gini coefficient based on post-tax post-social transfer monthly household income will increase.

(1) is incorrect. The rates concession mainly benefits people with high income. If the government cancels the rates concession, their income after taxes and social transfers will decrease, while people with low income will not be affected. The income gap between the rich and the poor will be narrowed and the Gini coefficient based on post-tax post-social transfer monthly household income will decrease.

23. Answer: D

Compared with Country B, Country A's Lorenz curve is further away from the line of perfect income equality. The area between the line of perfect income equality and the Lorenz curve is larger, so the Gini coefficient of Country A is higher.

Option A is incorrect. We cannot infer the tax systems of the two countries from the Lorenz curves based on pre-tax income.

Options B and C are incorrect. The information does not reflect the level of GDP or the living standard of people at different income levels.

24. Answer: C

GDP at factor cost = Private consumption expenditure + Gross investment + Government consumption expenditure + Net exports – Indirect taxes + Subsidies

$$1,400 = 500 + (220 + 80 - 10) + 500 + (280 - 100 - 60) - X + 50$$

$$X = 60$$

25. Answer: A

Implicit price deflator of GDP = (Nominal GDP / Real GDP) × 100

In year 1, the percentage change in nominal GDP is 1%, while the percentage change in the price level is 2%. Since the percentage increase in nominal GDP is smaller than the percentage increase in the price level, real GDP decreased in year 1.

In year 2, the percentage change in nominal GDP is 1%, while the percentage change in the price level is –3%. Since nominal GDP increased and the price level decreased, real GDP increased in year 2.

Option B is incorrect.

In year 1, since the percentage increase in nominal GDP (1%) is greater than the percentage increase in the population (0%), per capita nominal GDP increased.

In year 2, since the percentage increase in nominal GDP (1%) is smaller than the percentage increase in the population (3%), per capita nominal GDP decreased.

Option C is incorrect.

$$1 + \text{Growth rate of real GDP} = (1 + \text{Growth rate of nominal GDP}) / (1 + \text{Growth rate of price level})$$

In year 2:

$$1 + \text{Growth rate of real GDP} = (1 + 1\%) / [(1 + (-3\%))]$$

$$\text{Growth rate of real GDP} = 4\%$$

As the percentage increase in real GDP (4%) is larger than the percentage increase in population size (3%), per capita real GDP increased in year 2.

Option D is incorrect. There is insufficient information to determine whether the inflation rate in year 1 increased or decreased compared to the previous year.

26. Answer: A

The professor worked in Hong Kong for less than a year, so he/she is not a resident of Hong Kong.

Therefore, the salary he/she earned would be regarded as factor income earned by a non-resident within the economic territory and should be deducted from the GDP of Hong Kong to calculate its GNI.

Options B and C are incorrect. They would be regarded as factor income earned by residents outside the economic territory and should be added to the GDP of Hong Kong to calculate its GNI.

Option D is incorrect. The fee reflects the value of production of a resident producing unit and would be included in the GDP and GNI of Hong Kong.

27. Answer: B

When firms expect the general price level to increase, they will reduce their present supply. Short run aggregate supply will decrease.

Option A is incorrect. Abolishing tax incentives for firms will lead to a decrease in investment expenditure, causing aggregate demand to decrease without affecting short run aggregate supply.

Option C is incorrect. An increase in people's desire to save will lead to a decrease in private consumption expenditure, causing aggregate demand to decrease without affecting short run aggregate supply.

Option D is incorrect. When the price of raw materials decreases sharply, the cost of production will decrease. Short run aggregate supply will increase.

28. Answer: D

The proposal will increase the cost of visiting Hong Kong for tourists, causing a decrease in the number of tourists visiting Hong Kong. Exports of services and net exports will decrease. Aggregate demand will decrease.

Option A is incorrect. In Hong Kong, the hotel accommodation tax is an indirect tax.

Option B is incorrect. Although the proposal will increase government revenue, it will not necessarily lead to a fiscal surplus. Whether there is a fiscal surplus depends on the government's fiscal balance.

Option C is incorrect. As the tax takes a decreasing proportion of income as tax payment as income increases, the tax paid by taxpayers is not proportional to their income. Therefore, it is not in line with the equality principle of taxation put forth by Adam Smith.

29. Answer: A

When the central bank prints more money, the money supply will increase and the interest rate will decrease. Consumption and investment will increase. Aggregate demand will increase and the inflationary (output) gap will widen.

Option B is incorrect. When the government raises the business registration fee, investment expenditures will decrease. Aggregate demand will decrease and the inflationary (output) gap will be narrowed.

Option C is incorrect. The increase in the transport cost of raw materials will cause the short run aggregate supply to decrease, and the inflationary (output) gap will be narrowed.

Option D is incorrect. When the major trading partner of the economy experiences an economic recession, net exports will decrease. Aggregate demand will decrease and the inflationary (output) gap will be narrowed.

30. Answer: C

31. Answer: D

$M2 = M1 + \text{Customers' savings and time deposits with licensed banks} + \text{Negotiable certificates of deposit (NCDs) issued by licensed banks held outside the banking sector}$

When Mr. Chan withdraws HK\$10,000 from a time deposit account in a restricted licence bank and uses the amount to purchase an NCD issued by a licensed bank, this will increase M2 by HK\$10,000.

Option A is incorrect. When Miss Li withdraws HK\$10,000 from her time deposit account in a licensed bank and holds it in cash, M2 will not change.

Option B is incorrect. When Miss Wong deposits HK\$500,000 in cash into a time deposit account in a restricted licence bank, M2 will decrease.

Option C is incorrect. When Mr. Cheung transfers HK\$10,000 from his savings account in a licensed bank to his current account in the same bank, M2 will not change.

32. Answer: B

After deposit creation, the amount of deposits in the banking system is \$1,500.

$$\text{New excess reserves} = (\$500 - \$1,500 \times 25\%) = \$125$$

Option A is incorrect.

$$\text{Money supply} = \text{Cash in public circulation} + \text{Bank deposits} = \$50 + \$1,500 = \$1,550$$

Option C is incorrect. The reserves of the banking system remain unchanged at \$500.

Option D is incorrect.

$$\text{Monetary base} = \text{Cash in public circulation} + \text{Bank reserves} = \$50 + \$500 = \$550$$

33. Answer: B

$$\text{Reserves} = \text{Monetary base} - \text{Cash in public circulation} = \$150 \text{ million} - \$70 \text{ million} = \$80 \text{ million}$$

$$\text{Deposits} = \text{Reserves} \times \text{Actual banking multiplier} = \$80 \text{ million} \times 4 = \$320 \text{ million}$$

$$\text{Required reserves} = \text{Deposits} \times \text{Required reserve ratio} = \$320 \text{ million} \times 20\% = \$64 \text{ million}$$

$$\text{Excess reserves} = \text{Actual reserves} - \text{Required reserves} = \$80 \text{ million} - \$64 \text{ million} = \$16 \text{ million}$$

$$\text{Loans} = \text{Deposits} - \text{Reserves} = \$320 \text{ million} - \$80 \text{ million} = \$240 \text{ million}$$

$$\begin{aligned} \text{Money supply} &= \text{Cash in public circulation} + \text{Bank deposits} = \$70 \text{ million} + \$320 \text{ million} \\ &= \$390 \text{ million} \end{aligned}$$

34. Answer: D

$$\begin{aligned} \text{Real rate of return on holding money} &= \text{Nominal rate of return on holding money} - \text{Actual inflation rate} \\ &= 0\% - (-1\%) = 1\% \end{aligned}$$

Therefore, the real rate of return on holding money is greater than the nominal rate of return on holding money.

Option A is incorrect. Nominal rate of return of holding money = 0%

Option B is incorrect.

$$\text{Actual real interest rate} = \text{Nominal interest rate} - \text{Actual inflation rate} = 5\% - (-1\%) = 6\%$$

Option C is incorrect. Expected real interest rate = Nominal interest rate – Expected inflation rate

The question does not provide information on the expected inflation rate.

35. Answer: B

Quantity theory of money: $MV = PY$, where M = money supply, V = velocity of circulation of money, P = price level and Y = real output.

Assume V and Y are both constant.

When the central bank continues to sell government bonds to the public, the money supply (M) will decrease continuously, causing the price level (P) to decrease continuously, i.e. there will be deflation.

36. Answer: C

The labour force is the sum of the employed population and unemployed population (= 7,900,000).

$$\text{Unemployment rate} = (\text{Unemployed population} / \text{Labour force}) \times 100\%$$

$$= (400,000 / 7,900,000) \times 100\% = 5.06\%$$

Option B is incorrect. Employment rate = (Employed population / Labour force) $\times$ 100%

$$= (7,500,000 / 7,900,000) \times 100\% = 94.94\%$$

Option D is incorrect.

$$\text{Underemployment rate} = (\text{Underemployed population} / \text{Labour force}) \times 100\%$$

$$= (120,000 / 7,900,000) \times 100\% = 1.52\%$$

37. Answer: D

(2) is correct. The tax system shown in the diagram is a progressive tax. A progressive tax reduces income inequality as the average tax rate paid by high-income groups is higher than that of low-income groups. The distribution of income therefore becomes more even.

(3) is correct. The x-axis of the diagram shows the taxable income, which is the amount used to calculate the amount of tax payment after deducting tax allowances and tax deductions. Therefore, we cannot determine whether there is a tax allowance for the tax.

38. Answer: A

When the citizens stop using the electronic payment tools, the transaction demand for money will increase, causing the interest rate to increase. Consumption and investment will decrease. Aggregate demand will decrease and the output level will decrease.

Option A is correct. When more enterprises change from monthly payments to weekly payments, the transaction demand for money will decrease and the interest rate will decrease. Consumption and investment will increase, aggregate demand will increase and the output level will increase.

Option B is incorrect. Increasing the allowance for the salaries tax rate will raise people's disposable income, so private consumption expenditure will increase and aggregate demand will increase. The output level will increase. However, when people's disposable income increases, the transaction demand for money will increase and the interest rate will further increase.

Option C is incorrect. When the government issues more government bonds to the public, the money supply will decrease and the interest rate will increase. Consumption and investment will decrease. Aggregate demand will decrease and the output level will further decrease.

Option D is incorrect. When the risk of investing in the stock market increases, the asset demand for money will increase, causing the interest rate to further increase. Consumption and investment will decrease. Aggregate demand will decrease and the output level will further decrease.

39. Answer: A

When the government raises the standard rate of salaries tax, people's disposable income will decrease. Aggregate demand will decrease and the output level will decrease.

Raising the standard rate of salaries tax may reduce people's incentive to work, resulting in a fall in taxable income. The tax revenue from salaries tax may increase or decrease, depending on the relative extent of the decrease in tax revenue due to the decrease in people's taxable income and the increase in tax revenue due to the rise in the tax rate.

40. Answer: C

Holding more international competitions to attract tourists will increase exports of services. Aggregate demand will increase and aggregate output will increase.

When exports of services increase, the balance of invisible trade will improve and the balance of payments surplus will increase.

Option A is incorrect. Cancelling the tax incentives for foreign investors may lead to a reduction in their investments. Aggregate demand will decrease and aggregate output will decrease.

When foreign investors reduce their investment, capital inflow will decrease. The balance of payments surplus will decrease.

Option B is incorrect. When the government increases the amount of transfer payments, people's disposable income will increase. Private consumption expenditure will increase and aggregate demand will increase. Aggregate output will increase.

When people's disposable income increases, their demand for imports will increase. The balance of visible trade will worsen and the balance of payments surplus will decrease.

Option D is incorrect. Importing more raw materials from foreign countries will lead to an increase in imports of goods. Aggregate demand will decrease and aggregate output will decrease.

When imports of goods increase, the balance of visible trade will worsen. The balance of payments surplus will decrease.

41. Answer: D

Increasing the discount rate will cause the money supply to decrease, and the money supply curve will shift to the left to M_{s1} .

When the liquidity of other interest-bearing assets increases, the asset demand for money will decrease. The money demand curve will shift to the left to M_{d1} .

42. Answer: C

Country A's opportunity cost of producing 1 unit of clothing = 2 (= 10/5) units of toys

Country B's opportunity cost of producing 1 unit of clothing = 5 (= 20/4) units of toys

Since the opportunity cost of producing 1 unit of clothing in Country A is lower than that in Country B, Country A has a comparative advantage in producing clothing. It will specialise in producing clothing and export clothing.

Terms of trade: 1 unit of clothing = 2.5 units of toys

Gain per unit of export for the clothing-exporting country (i.e. Country A) = 2.5 units of toys – 2 units of toys – 0.1 units of toys (transportation cost) = 0.4 units of toys

Total gain from trade for the clothing-exporting country = 0.4 units of toys $\times$ 4 = 1.6 units of toys

43. Answer: C

When the world price decreases, the import volume of the good will remain unchanged and equal to the quota limit. The domestic price of the good will remain at P_1 . The domestic production will remain unchanged at Q_2 and the revenue of domestic producers will remain unchanged.

Option A is incorrect. The domestic price of the good will remain unchanged at P_1 .

Option B is incorrect. The import volume will remain unchanged at the quota limit, i.e. $(Q_3 - Q_2)$.

Option D is incorrect. The original total import value is $P_w \times (Q_3 - Q_2)$. As the world price decreases while the import volume remains unchanged at the quota limit, the total import value will decrease, i.e. be smaller than $P_w \times (Q_3 - Q_2)$.

44. Answer: B

(1) is correct.

Current account balance = Visible trade balance + Invisible trade balance + Net factor income from abroad + Net current transfers

= $[230 + 170 + (40 - 20) + (20 - 25)]$ million = \$415 million

There is a surplus of \$415 million in the current account.

(3) is correct. The balance of payments reflects the net change in reserve assets. Reserve assets decrease by \$50 million, meaning that there is a balance of payments deficit of \$50 million.

(2) is incorrect.

GNI = GDP + Net factor income from abroad

As the value of net factor income from abroad is positive, the economy's GNI is higher than its GDP.

45. Answer: D

The Hong Kong dollar appreciates against the Japanese yen. Given the price of Hong Kong goods remains unchanged, Hong Kong's exports will become more expensive. Japanese importers will buy fewer Hong Kong goods. Demand for Hong Kong's exports will decrease and the quantity transacted will decrease. The total value of Hong Kong's exports to Japan (in terms of HKD) will decrease.

Option A is incorrect. As the Hong Kong dollar appreciates against the Japanese yen, the price of Japanese goods and services (in terms of HKD) will decrease. The cost of travelling to Japan (in terms of HKD) will decrease and so more Hong Kong tourists will visit Japan.

Option B is incorrect. The price of goods exported from Japan to Hong Kong will decrease (in terms of HKD), resulting in an increase in the quantity of Japanese goods exported to Hong Kong. The total value of Japanese goods exported to Hong Kong (in terms of HKD) may increase or decrease, depending on the price elasticity of demand.

Option C is incorrect. The price of Hong Kong's exports to Japan will increase (in terms of JPY), so the competitiveness of Hong Kong goods exported to Japan will decrease.

Paper 2

Section A (44 marks)

1. (a) (i) Economic goods. (1 mark)
This is because the designated bags are produced from scarce resources which have alternative uses. (1 mark)
- (ii) Private goods. (1 mark)
The designated bags are rival in consumption. When a person gets a bag, the number of bags available for others will decrease. (2 marks)
- OR
The designated bags are excludable in consumption. When a person uses a bag, he/she can exclude others from using it. (2 marks)
- (b) Positive statement. (1 mark)
It is refutable by facts. (1 mark)
2. (a) An advantage:
- Joey and Queenie can enjoy limited liability.
- The fruit shop can enjoy better continuity.
- Any reasonable answer
(Mark the **FIRST** point only) } 1@, max: 1 mark
- A disadvantage:
- The fruit shop is subject to a higher profits tax rate.
- Any reasonable answer
(Mark the **FIRST** point only) } 1@, max: 1 mark
- (b) Imports of goods would increase. This would reduce Hong Kong's current account surplus (or enlarge its deficit). (2 marks)
- (c) Reasons:
- Lower buying costs: The fruit shop may be given discounts when it purchases raw materials in large quantities, resulting in a lower average cost.
- Lower cost of borrowing: The fruit shop may have more assets that can be used as collateral. It will be generally easier and cheaper for it to borrow money from banks. This reduces interest expenses and thus lowers average cost.
- Wider spread of marketing expenses: The fruit shop can spread the costs of marketing, such as advertisement expenditures, over a larger quantity of output. This lowers average cost.
- Any reasonable answers
(Mark the **FIRST TWO** points only) } 1@, max: 2 marks

3. Verbal elaboration:

When Country A's economy experienced a recession, people's income would decrease and the demand for durians would decrease. The quantity transacted would decrease. In the case of imposing a per-unit tax, the decrease in demand would not affect the difference between marginal benefit and marginal cost, so the deadweight loss in the market of durians would not change. (2 marks)

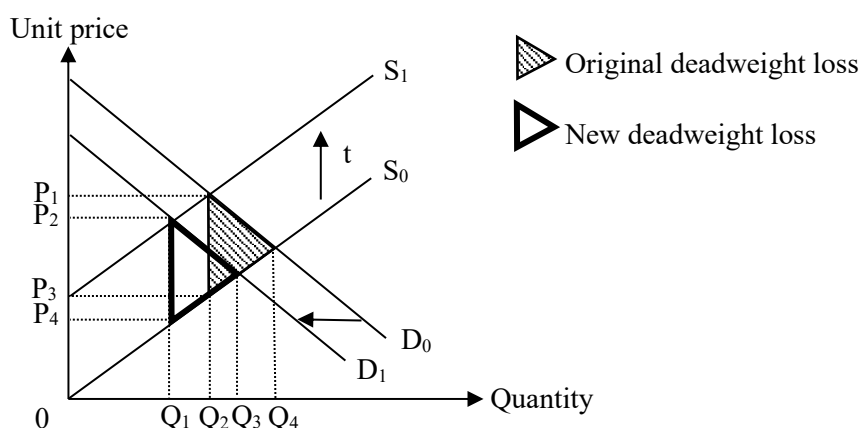
Illustrate in the diagram:

- leftward shift of demand curve (1 mark)

- price decreases (from P_1 to P_2), quantity transacted decreases (from Q_2 to Q_1) (1 mark)

- correct positions of original deadweight loss and new deadweight loss (1 mark)

- original deadweight loss = new deadweight loss (1 mark)



4. (a) Recession (Contraction). (1 mark)

(b) Reasons:

- Although the real GDP of Country A decreased, the population might have also decreased. If the per capita real GDP increased, the living standard might have increased.
- Country A's income distribution might have become more even.
- People in Country A might have enjoyed more leisure time.
- Any reasonable answers

1@, max: 2 marks

(Mark the **FIRST TWO** points only)

(c) No. (1 mark)

Unemployment rate = (Unemployed population / Labour force) $\times$ 100% (1 mark)

The increase in the unemployment rate may be caused by a decrease in the labour force while the unemployed population remained unchanged. (2 marks)

OR

Any reasonable answer (2 marks)

5. (a) Inflation rate = $[(105.6 - 101.3) / 101.3] \times 100\% = 4.24\%$ (1 mark)

- (b) QTM: $MV = PY$, where M = money supply, V = velocity of circulation of money, P = price level and Y = real output, assuming V is constant.

When the money supply (M) decreases, if the percentage decrease in M is smaller than the percentage decrease in Y , P will increase. (3 marks)

- (c) Mr. Chan would lose because the real value (purchasing power) of his monetary assets would decrease. (2 marks)

6. (a) They are a direct tax as the tax burden cannot be shifted to other parties. (2 marks)

- (b) Equity: The new tax is no longer proportional, meaning that it is inconsistent with the equity principle.

Economy: The new tax system may incur higher administration costs, making it less in line with the economy principle.

Certainty: Both the new and old tax systems are equally well defined and announced publicly, so there is no difference between the new and old tax systems in terms of the certainty principle.

Convenience: There is no change in the method of tax collection, so there is no difference between the new and old tax systems in terms of the convenience principle.

2@, max: 2 marks

(Mark the **FIRST** point only)

- (c) A possible undesirable effect:

- An increase in the income tax rate may lead to a decrease in private consumption expenditure, and aggregate demand may decrease.

Therefore, aggregate output may decrease.

- When the income tax rate increases, workers' incentive to work may be lower and the labour supply may decrease.

- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

7. (a) Country A's opportunity cost of producing 1 unit of coffee = 1.25 (= 50 / 40) units of clothing (1 mark)

Country B's opportunity cost of producing 1 unit of coffee = 2 (= 20 / 10) units of clothing (1 mark)

As Country A has a lower opportunity cost in producing coffee than Country B, Country A will export coffee. (1 mark)

- (b) Country A's gain from exporting 1 unit of coffee = 1.5 units of clothing – 1.25 units of clothing – 0.1 units of clothing = 0.15 units of clothing

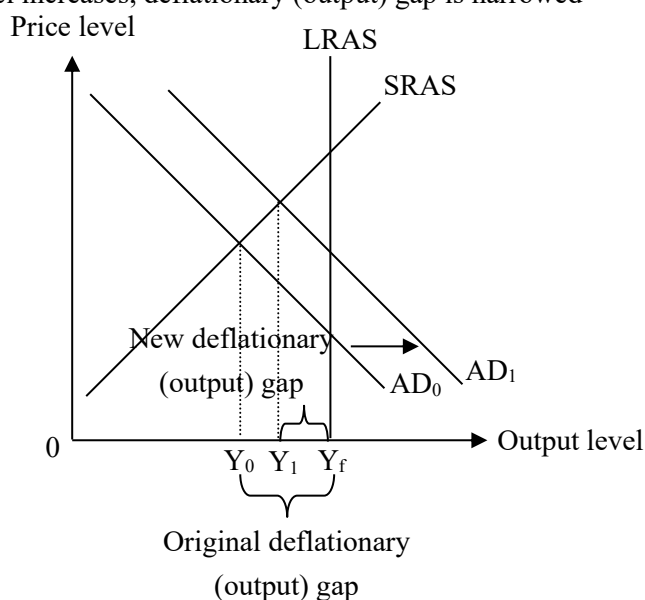
Therefore, Country A's gain from exporting 10 units of coffee = 1.5 (= 0.15 × 10) units of clothing. (3 marks)

Section B (60 marks)

8. (a) Tertiary production. (1 mark)
 (b) No. (1 mark)
 No production is involved. (1 mark)
 (c) Verbal elaboration:
 The disposable income of the students will increase and private consumption expenditure will increase. (2 marks)
 Aggregate demand will increase. (1 mark)
 The output level will increase, narrowing the deflationary (output) gap. (1 mark)

Illustrate in the diagram:

- correct position of original deflationary (output) gap (1 mark)
- rightward shift of AD curve (1 mark)
- output level increases, deflationary (output) gap is narrowed (1 mark)



- (d) No. (1 mark)
 The increase in the amounts of the University Admission Scholarships would increase the value of the third option. If the value of the third option increased to a level higher than that of the second option, enrolling in an accounting programme at the Chinese University of Hong Kong would become the highest-valued option forgone. The cost of choosing to enroll in an arts programme abroad would change from the value of enrolling in a law programme at the University of Hong Kong to the value of enrolling in an accounting programme at the Chinese University of Hong Kong. (2 marks)
9. (a) (i) Negative externality. Residents of Santorini suffer from an overcrowded environment due to too many tourists. However, the residents do not receive any compensation. There is an external cost. The marginal social cost is higher than the marginal social benefit. The quantity of tourists is higher than the efficient level. (3 marks)
 (ii) When the cruise passenger fees are imposed, the marginal private cost of visiting Santorini will increase. The gap between marginal private cost and marginal social cost will be narrowed. The equilibrium quantity of tourists will decrease towards the socially optimal level. (2 marks)

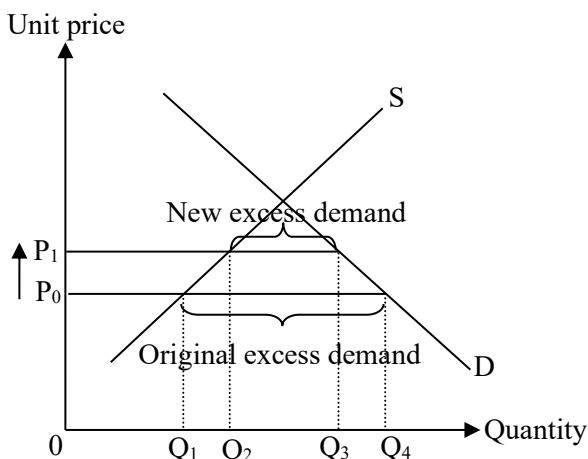
- (b) Both long-distance tourists and short-distance tourists need to pay the same cruise passenger fees. After adding the fees, the relative price of visiting Santorini for long-distance tourists (relative to short-distance tourists) will decrease. According to the law of demand, more long-distance tourists will visit Santorini relative to short-distance tourists. The proportion of long-distance tourists (e.g. Asian visitors) to Santorini will increase, while the proportion of short-distance tourists (e.g. European visitors) will decrease. (4 marks)
- (c) Charging cruise passenger fees will cause the price of visiting Santorini during the peak season to increase, but the number of tourists may not decrease significantly, meaning that the quantity demanded may not decrease significantly. In this case, the percentage decrease in the quantity demanded will be smaller than the percentage increase in price, showing that the demand for trips to Santorini during the peak season is inelastic. (3 marks)
- (d) The cost of monitoring the service quality of tour guides is lower and the incentive for tour guides to work harder and provide higher-quality service to customers is higher. (2 marks)

10. (a) (i) Verbal elaboration:

The initial taxi fare was below the equilibrium level and there was a shortage of/an excess demand for taxi services during peak hours. When the taxi fare increases, the quantity supplied of taxi services will increase while the quantity demanded of taxi services will decrease. The excess demand for taxi services will decrease and the waiting time for a taxi will be reduced. (3 marks)

Illustrate in the diagram:

- price increases (1 mark)
- correct position of original excess demand (1 mark)
- smaller excess demand (1 mark)



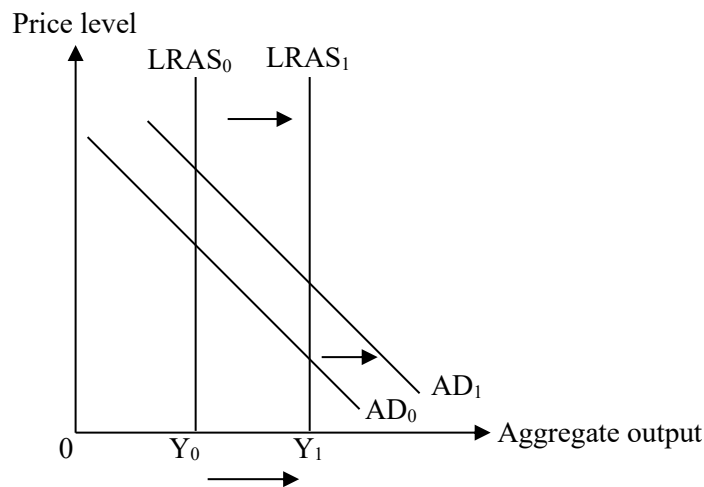
- (ii) No. The full cost of people using taxi services includes monetary cost and time cost. Given that the waiting time for a taxi is reduced, if the time cost saved is higher than the additional monetary cost, people's cost of using taxi services will decrease after the increase in taxi fares. (3 marks)

(b) Verbal elaboration:

When the government introduces the taxi fleet regime and launches the Smart Traffic Fund, investment expenditure will increase and aggregate demand will increase. At the same time, the Smart Traffic Fund will encourage enterprises and organisations to research and apply vehicle-related innovations and technology. This will increase Hong Kong's productive capacity, resulting in an increase in long run aggregate supply. In the long run, aggregate output will increase. (4 marks)

Illustrate in the diagram:

- rightward shift of AD curve (1 mark)
- rightward shift of LRAS curve (1 mark)
- higher aggregate output in the long run (1 mark)



- (c) When online ride-hailing platforms are legalised, working as a driver for these platforms will also become legal. At the same time, more people are expected to use these online ride-hailing services due to the legalisation. The income for drivers on these platforms is thus expected to increase. People will be more willing to switch occupations and become drivers on these platforms, so the labour mobility of Hong Kong will increase. (3 marks)

(d) Marks awarded for effective communication (EC: max 2 marks)

Marks	Performance
2	• Arguments supported with the source/data and appropriate economic theories • Relevant materials presented • Well-organised and coherent answers presented without repetition of ideas • Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
1	• Arguments presented with some support of the source/data and economic theories • Some irrelevant materials presented • Answers presented in a less organised way with some repetition of ideas • A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	• Arguments presented with no support of the source/data and economic theories • Materials unrelated to the gist of the question presented • Inconsistent arguments presented • Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 12 marks. Answers may include the following:

Objective 1:

- Suggestion I will increase the number of vehicles on online ride-hailing platforms. When the supply of online ride-hailing services increases, the price will decrease and the quantity transacted will increase. The total revenue of the online ride-hailing services market will increase if the demand for the services is elastic.
- For Suggestion I, legalising online ride-hailing platforms may increase people's confidence in online ride-hailing services, leading to an increase in the demand. Lowering the requirements for hire car permits will increase the supply of ride-hailing services. The quantity transacted will increase. The change in price will be uncertain. If the effect of the increase in demand on price is greater than the effect of the increase in supply on price, then the price will increase. As a result, the total revenue of the online ride-hailing services market will increase.
- For Suggestion II, after taxis upgrade their vehicle equipment, it will become more convenient for people to use taxi services, and demand for taxi services will increase, with reference to Miss Lee's view in Source B. Therefore, the total revenue of the taxi services market will increase.
- Both suggestions may achieve the objective of increasing the total revenue in the targeted market. The government will prefer Suggestion II if it prefers a more certain effect.

Objective 2:

- Suggestion I will increase the number of vehicles on online ride-hailing platforms. More people can become drivers on these platforms and their income may increase. The government's tax revenue from the salaries tax may increase, which is favourable to the government's fiscal balance.
- For Suggestion II, providing tax deductions for taxis' expenditures related to upgrading vehicle equipment will reduce government tax revenue. On the other hand, demand for taxi services may increase after the taxis upgrade their vehicle equipment. The income of taxi drivers may increase, and so government tax revenue from the salaries tax may increase. Therefore, the overall effect on the government's fiscal balance will be uncertain.
- The government will prefer Suggestion I, which can better achieve the objective of improving the government's fiscal balance.

Objective 3:

- Suggestion I will increase the number of job positions in Hong Kong. If some of the unemployed become drivers on these platforms, the unemployment rate will decrease.
- In Hong Kong, the government has set a limit on the number of taxi licences. For Suggestion II, if the number of taxi licences remains unchanged, the number of taxi drivers required may not change significantly. Therefore, Suggestion II may have no effect on the unemployment rate.
- The government will prefer Suggestion I, which can better achieve the objective of lowering the unemployment rate.

Section C (16 marks)

11. (a) Illustrate in the diagram:

- correct position of Q_C (1 mark)

- correct position of P_C (1 mark)

(b) Illustrate in the diagram:

- correct position of MR curve (1 mark)

- correct position of Q_M (1 mark)

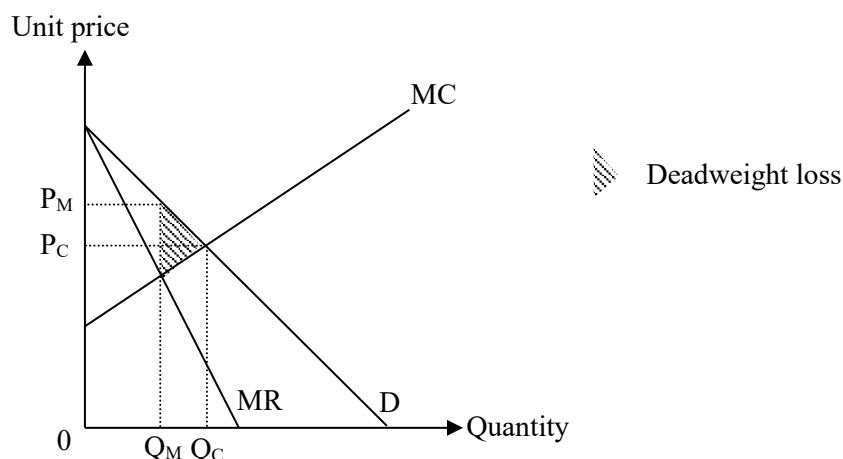
- correct position of P_M (1 mark)

(c) Verbal elaboration:

Before the change in the market structure, at the profit-maximising output (Q_C), marginal benefit equals marginal cost. After the change in the market structure, at the profit-maximising output (Q_M), marginal benefit is greater than marginal cost. There is a deadweight loss and economic efficiency is worsened. (3 marks)

Illustrate in the diagram:

- correct position of the deadweight loss (1 mark)



(d) (i) Price discrimination is a pricing behaviour whereby a seller charges different prices for the same good or service to different customers, in which the differences cannot be explained by differences in costs. In this case, the same good produced at the same cost (seafood) is sold at different prices to different groups of customers (customers aged 70 or above and customers aged below 70). (3 marks)

(ii) A condition:

- The seller is a price searcher. It can charge its own price without losing all its customers.

- The market is separable.

- Any reasonable answer

(Mark the **FIRST** point only)

1@, max: 1 mark

(e) Tie-in sales.

(1 mark)

Possible harms to consumers:

- The market price of delivery service is higher.

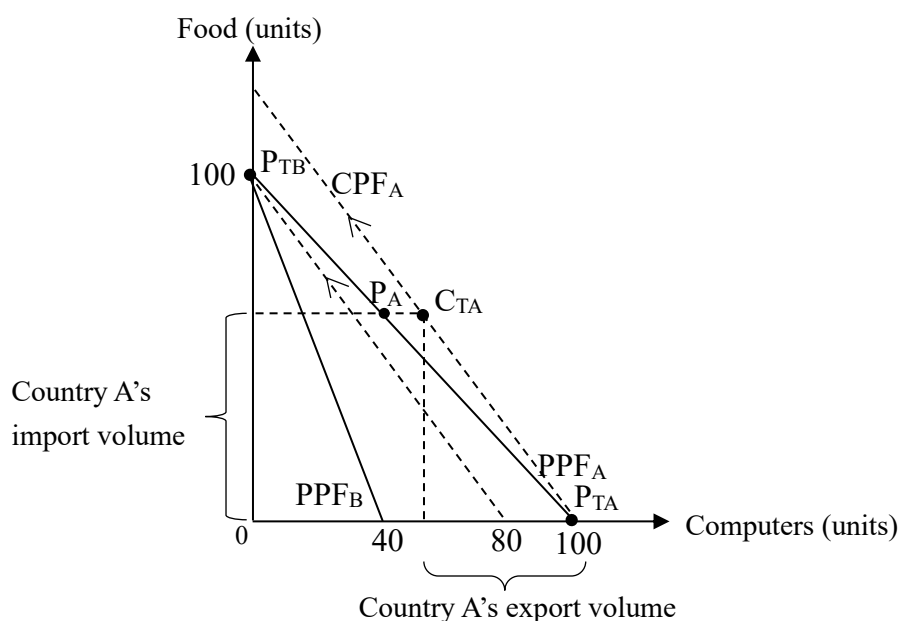
- Consumers lack choices.

- Any reasonable answers

(Mark the **FIRST TWO** points only)

1@, max: 2 marks

12. (a) Illustrate in the diagram:
 - correct position of Country B's PPF (PPF_B) (1 mark)
- (b) Country A has an absolute advantage in producing computers, (1 mark)
 because it can produce more computers than Country B using the same amount of resources. (1 mark)
- (c) (i) Illustrate in the diagram:
 - correct position of P_{TA} (1 mark)
 - correct position of P_{TB} (1 mark)
- (ii) Illustrate in the diagram:
 - correct position of CPF_A (1 mark)
 - CPF_A parallel to the world price line (the dotted line) (1 mark)
- (iii) Illustrate in the diagram:
 - correct position of C_{TA} (1 mark)
 - correct position of export volume (1 mark)
 - correct position of import volume (1 mark)



- (d) No. As the opportunity costs of production in Country A and Country B do not change, Country A still has a comparative advantage in producing computers. Thus, the direction of trade between the two countries will not be affected. (2 marks)
- (e) Economic growth of Country A will improve after engaging in trade because increased production, business activities, and employment opportunities in the import/export industry will lead to higher per capita real GDP. However, more people may leave school earlier to participate in trading activities, decreasing the mean years of schooling / pollution may become more serious as production activities increase after trade, lowering the life expectancy at birth of the people of Country A. Therefore, economic development of the country may worsen. (4 marks)